

Sanjaya Maharjan

AI & Cloud Architect

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SUMMARY

AI and cloud architect with fourteen years of production engineering since 2012. I build AI agents and LLM applications, including an agentic corporate-footprint mapper for Monotype, and the platforms they run on: .NET microservices and secure APIs, React and Angular front-ends, and AWS infrastructure defined in Terraform, delivered for Monotype, HP, Deloitte and the State of Colorado. I own the architecture, write the code, and stay for production.

AI & ML WORK

- **Corporate Footprint Mapper** (built for Monotype, private): an agentic tool that maps a company's legal entities from the ultimate parent to the nth subsidiary and attaches each entity's brands, websites, apps and digital assets, with web scraping built in.
- **Agent organization** (R&D, private): role-separated agents on Claude Code — CEO, Strategist, Researcher, Builder, Reviewer, Growth, Comms, Legal and Investor. Every claim needs a cited source and a Reviewer can block any plan; memory in files and git keeps every decision auditable, and scheduled cloud runs keep it working. 606 commits and 145 reviews over 89 active days (counted September 2026).
- **ML in client work:** ML-driven font impression and emotion analysis for Fontworks' LETS and Monotype's Research 360.
- **Open source:** Fake-Image-Detector, which detects manipulated images by combining ML with analytical signals (sensor noise, FFT/DCT spectra, resampling, lighting physics), with ONNX inference, a model registry and drift detection; scrape-diagnose — a CLI that explains scraper 403/429/empty-HTML failures; json-typed-code — a VS Code extension that turns JSON into TypeScript, Python, Go or Rust types.
- **R&D:** IncidentAtlas (reliability agent with CockroachDB vector-indexed memory, CockroachDB × AWS agentic-memory challenge); H-VF Engine (VAE-generated variable fonts with CNN scoring); Disaster Damage Assessment (UNet and Siamese change detection, Grad-CAM); D-CAT (reinforcement-learning adaptive typography); air-quality and public-health forecasting (Prophet, SARIMA, LSTM, SEIR); IntelligentResume (Llama 3.1 via Ollama with guardrails); Trace the Ace (DrivenData transcript classifier).

EXPERIENCE

Lead Software Engineer — UBA Solutions (Monotype partner), Kathmandu

2024 – Present

- Own architecture for Research 360, which turns raw font metadata into a structured analysis and licensing platform.
- Set scalable deployment strategies for high-traffic font services (LETS, Fontplus), improving reliability and release safety.
- ML-driven font impression and emotion analysis; full-stack .NET, React and AWS, with performance gains from loading and data-throughput optimizations.
- Set up Cloudflare edge delivery and bot protection, with monitoring and alerting.

Senior Software Engineer — HP Inc., Texas

2024 – 2025

- Owned microservices design and AWS delivery (ECS, RDS, Lambda, VPC) for the HP Workforce Experience Platform.
- Architected secure REST APIs (OpenAPI/Swagger) with OAuth/JWT and secrets management; automated Terraform and Azure DevOps CI/CD.
- Improved React performance with lazy loading and route preloading, and built reusable TypeScript components.

Senior Software Engineer — Deloitte, Colorado

2023 – 2024

- Led API platform delivery for the State of Colorado's unemployment insurance platform: OAuth and API-key security, traffic management, analytics.
- Built workflow microservices and API Gateway integration with the IBM content engine on .NET Core; implemented SQS-backed async workloads and Redis caching.
- Delivered document and profile management in Angular 18 and React, with SSO, high availability and Azure DevOps CI/CD.

Senior Software Engineer — Dwaith Infotech, Minnesota (remote)

May – Sep 2023

- Modernized a legacy architecture onto .NET microservices handling 1M+ requests a day; React and Redux with TypeScript UI patterns.
- Optimized SQL and NoSQL stores (PostgreSQL, MySQL, MongoDB, DynamoDB) and containerized deployments; added Redis caching, async processing and monitoring.

Solution Architect / Lead Software Engineer — UBA Solutions (Monotype partner), Nepal

2018 – 2023

- Scaled and modernized MyFonts, Fonts.com, Linotype and FontShop on AWS with EC2 autoscaling, load balancers, RDS read replicas and S3.
- Migrated on-prem systems to AWS with Terraform; CI/CD with Jenkins and Octopus; improved monitoring and incident response.
- Strengthened security with OAuth/JWT, SSO, 3D Secure, CyberSource and Cloudflare bot protection; ran Docker and Kubernetes; optimized caching and CDN.

Lead Software Engineer — Novelty Technology (North Carolina, Nepal office)

2017 – 2018

- Led a microservices transition and full-stack delivery (Node.js, Angular, RxJS, ASP.NET Core) for CollegeRecon and Alithias healthcare analytics.

Senior Software Engineer — Grafi Offshore, Nepal

2014 – 2016

- Owned infrastructure and software design; PayPal and Adyen integrations for Lesling.com, a Japanese tobacco inventory system and Banaunu.com.

Software Engineer — Hatra Inc. and Channakya Software, Nepal

2012 – 2014

- Built a financial risk-management assistant, a Bank of Canada survey platform and the Subisu inventory system in C#, ASP.NET, Web API, Entity Framework and MSSQL.

SKILLS

- **AI & ML:** LLM apps and RAG, agents and tool use, vector search and agent memory, scraping and entity mapping, AWS Bedrock and SageMaker, Azure OpenAI, OpenAI and Anthropic APIs, ONNX and drift detection, Ollama and Hugging Face, spaCy, UNet/Siamese/VAE, Prophet/SARIMA/LSTM.
- **Backend:** C#, .NET and ASP.NET Core, Web API, REST and OpenAPI, Entity Framework and LINQ, Node.js and NestJS, Python and FastAPI.
- **Frontend:** React, Angular, Next.js, TypeScript, RxJS, Redux.
- **Cloud & DevOps:** AWS (ECS/ECR, EC2, Lambda, RDS, S3, VPC, SQS, CloudWatch, Route 53), Terraform, Docker, Kubernetes, Jenkins, Octopus, Azure DevOps, Cloudflare.
- **Data:** SQL Server, PostgreSQL, MySQL, MongoDB, DynamoDB, Redis.

EDUCATION & CERTIFICATION

- AWS Certified Developer — Associate (Amazon Web Services)
- Master's in Computer Science — Nepal College of Information and Technology
- Bachelor's in Computer Engineering — Khwopa Engineering College, Nepal